

Chapter 1:

Introduction to the World of Computers

Learning Objectives

1. Explain why it is essential to learn about computers today and discuss several ways computers are integrated into our business and personal lives.
2. Define a computer and describe its primary operations.
3. Identify the major parts of a personal computer, including input, processing, output, storage, and communications hardware.
4. Define software and understand how it is used to instruct the computer what to do.
5. Six basic categories of computers:

Learning Objectives

7. Explain What is meant by Number Systems.
8. Explain Basics of Binary system, converting to decimal and different operations.
9. Viewing Octal and Hexadecimal Systems.
10. Identify the major parts of a flowchart, how to create a flowchart to solve a given problem.

Learning Objectives

10. Explain what a network, the Internet, and the World Wide Web are, as well as how computers, people, and Web pages are identified on the Internet.
11. Describe how to access a Web page and navigate through a Web site.
12. Discuss the societal impact of computers, including some benefits and risks related to their prominence in our society.
13. Artificial Intelligence, IOT.

Overview

- This chapter covers:
 - What computers are, how they work, and how they are used
 - Computer terminology
 - An overview of the history of computers
 - The basic types of computers in use today

Computers in Your Life

- Before 1980
 - Computers were large, expensive
 - Very few people had access to them
 - Computers were mostly used for high-volume processing tasks
- Microcomputers in the early 80s
 - Inexpensive personal computers
 - Computer use increased dramatically
- Today
 - More than 80% of people households include a computer, and most use computers at work

Computers in the Home

- Computers used for a variety of tasks:
 - Looking up information and news
 - Exchanging e-mail
 - Shopping and paying bills
 - Watching TV and videos
 - Downloading music and movies
 - Organizing digital photographs
 - Playing games

Computers in the Home

- Wireless networking
 - Computers can be used in nearly any location
- Smart appliances
 - Traditional appliances with built-in computer or communication technology
- Smart homes
 - Household tasks are monitored and controlled by a main computer in the house

Computers in the Home



REFERENCE AND COMMUNICATIONS

Many individuals today have access to the Internet at home; retrieving information, obtaining news, viewing recipes, shopping online, and exchanging e-mail are popular home computer activities.



PRODUCTIVITY

Home computers are frequently used for editing and managing digital photos and home videos, creating and editing work-related documents, and other productivity tasks.



ENTERTAINMENT

Home computers and gaming consoles are becoming a central hub for entertainment, such as the delivery of photos, videos, music, games, and recorded TV.



FIGURE 1-2

Computer use at home.

Computers in Education

- Many students today have access to computers either in a classroom or a computer lab
- Colleges and universities are even more integrated
 - Wireless hotspots allow usage of personal laptops to connect to the college network
 - Some colleges require a computer for enrollment
- Distance learning
 - Students participate from locations other than the traditional classroom setting using computers and Internet access

Computers in Education



COMPUTER LABS AND CLASSROOMS

Many schools today have computers available in a lab or the library, as well as computers or Internet connections in classrooms, for student use.



CAMPUS WIRELESS HOTSPOTS

Many college students can access the Internet from anywhere on campus to do research, check e-mail, and more, via a campus hotspot.



DISTANCE LEARNING

With distance learning, students—such as these U.S. Army soldiers—can take classes from home or wherever they happen to be at the moment.



FIGURE 1-3

Computer use in education.

Computers on the Job

- Computers have become a universal on-the-job tool for decision-making, productivity, and communication
 - Used by all types of employees
 - Used for access control and other security measures
 - Use by service professionals is growing
 - Used extensively by the military
 - Employees in all lines of work need to continually refresh their computer skills

Computers on the Job



DECISION MAKING

Many individuals today use a computer to help them make on-the-job decisions.



PRODUCTIVITY

Many individuals today use a computer to perform on-the-job tasks efficiently and accurately.

FIGURE 1-4
Computer use on the job.



OFFSITE COMMUNICATIONS

Many individuals use portable computers or mobile devices to record data, access data, or communicate with others when they are out of the office.



AUTHENTICATION

Many individuals use authentication systems to punch in and out of work, access facilities, log on to company computers, or perform other security-related tasks.

Computers on the Go

- Computers are encountered in nearly every aspect of daily life
 - Consumer kiosks
 - ATM transactions
 - POS systems at retail stores
 - Portable computers or mobile devices
 - E-commerce systems
 - GPS systems

Computers on the Go



PORTABLE COMPUTERS

Many people today carry a portable computer or mobile device with them at all times or when they travel in order to remain in touch with others and Internet resources.



CONSUMER KIOSKS

Electronic kiosks are widely available to view conference or gift registry information, print photographs, order products or services, and more.



M-COMMERCE SYSTEMS

Allow individuals to pay for purchases using a mobile phone or other device.



CONSUMER AUTHENTICATION SYSTEMS

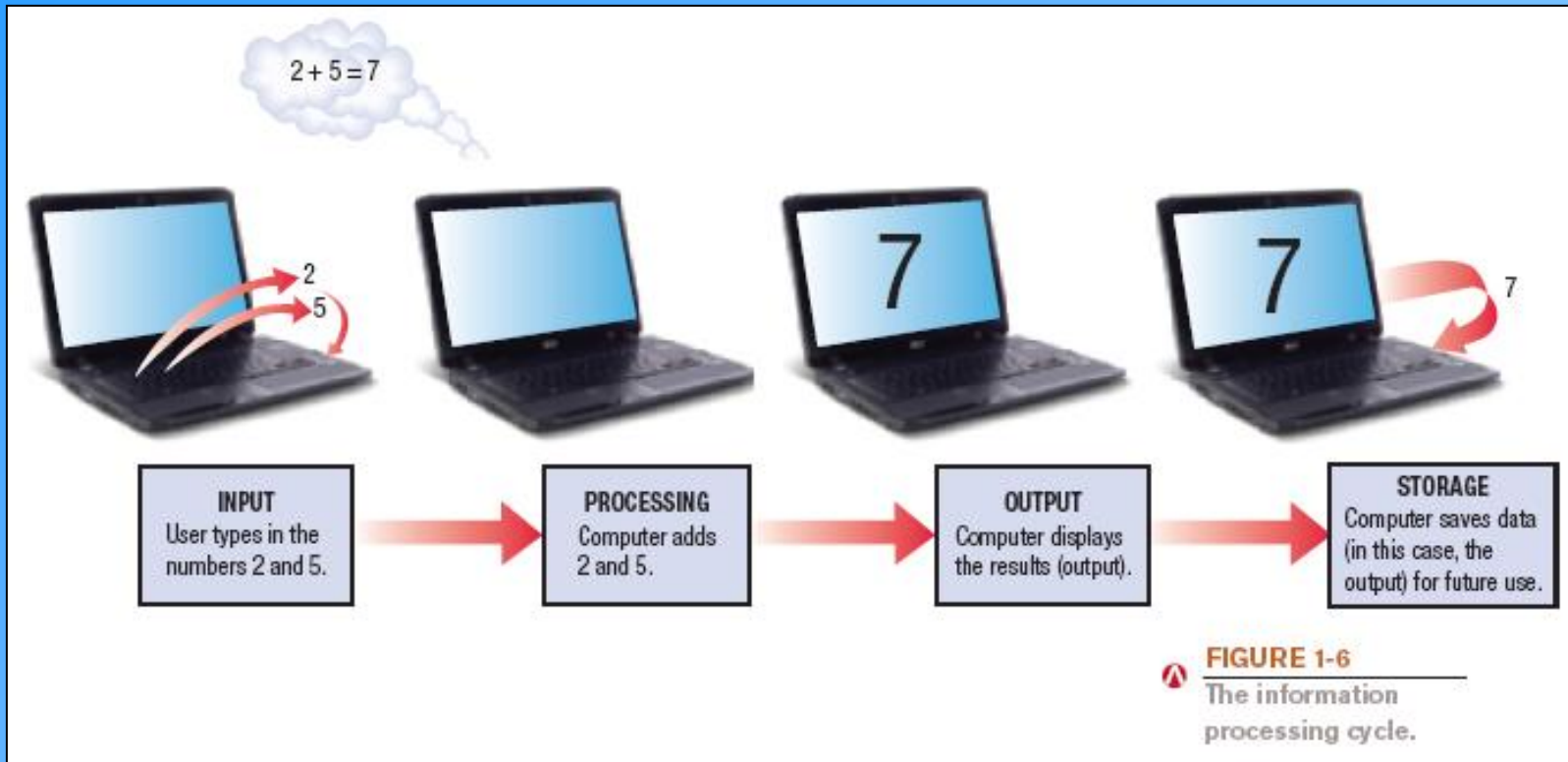
Allow access to facilities for authorized members only, such as for theme park annual pass holders, as shown here.

FIGURE 1-5
Computer use while on the go.

What Is a Computer and What Does It Do?

- Computer: A programmable, electronic device that accepts data, performs operations on that data, and stores the data or results as needed
 - Computers follow instructions, called programs, which determine the tasks the computer will perform
- Basic operations
 - Input: Entering data into the computer
 - Processing: Performing operations on the data
 - Output: Presenting the results
 - Storage: Saving data, programs, or output for future use
 - Communications: Sending or receiving data

What Is a Computer and What Does It Do?



Data vs. Information

- Data
 - Raw, unorganized facts
 - Can be in the form of text, graphics, audio, or video
- Information
 - Data that has been processed into a meaningful form
- Information processing
 - Converting data into information

Hardware

- Hardware: The physical parts of a computer
 - Internal hardware
 - Located inside the main box (system unit) of the computer
 - External hardware
 - Located outside the system unit
 - Connect to the computer via a wired or wireless connection
 - There is hardware associated with all five computer operations

Hardware



FIGURE 1-8

Common hardware
listed by operation.

INPUT	PROCESSING
Keyboard	CPU
Mouse	OUTPUT
Microphone	Monitor
Scanner	Printer
Digital camera	Speakers
Digital pen/stylus	Headphones/headsets
Touch pad/touch screen	Data projector
Joystick	STORAGE
Fingerprint reader	Hard drive
COMMUNICATIONS	CD/DVD disc
Modem	CD/DVD drive
Network adapter	Flash memory card
	Flash memory card reader
	USB flash drive

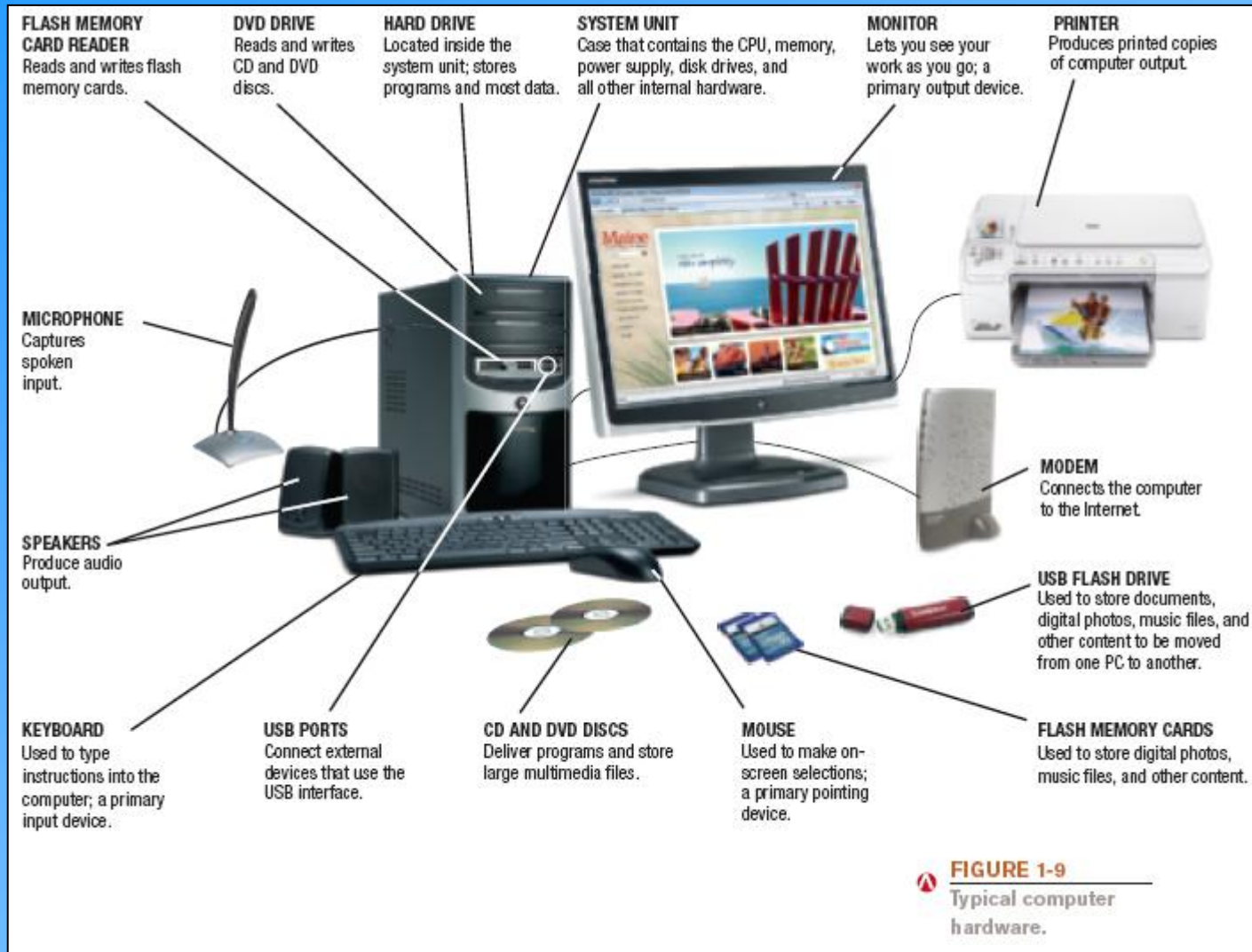
Hardware

- Input devices
 - Used to input data into the computer
 - Keyboards, mice, scanners, cameras, microphones, joysticks, touch pads, touch screens, fingerprint readers, etc.
- Processing devices
 - Perform calculations and control computer's operation
 - Central processing unit (CPU) and memory
- Output devices
 - Present results to the user
 - Monitors, printers, speakers, projectors, etc.

Hardware

- Storage devices
 - Used to store data on or access data from storage media
 - Hard drives, CD/DVD discs and drives, USB flash drives, etc.
- Communications devices
 - Allow users to communicate with others and to electronically access remote information
 - Modems, network adapters, etc.

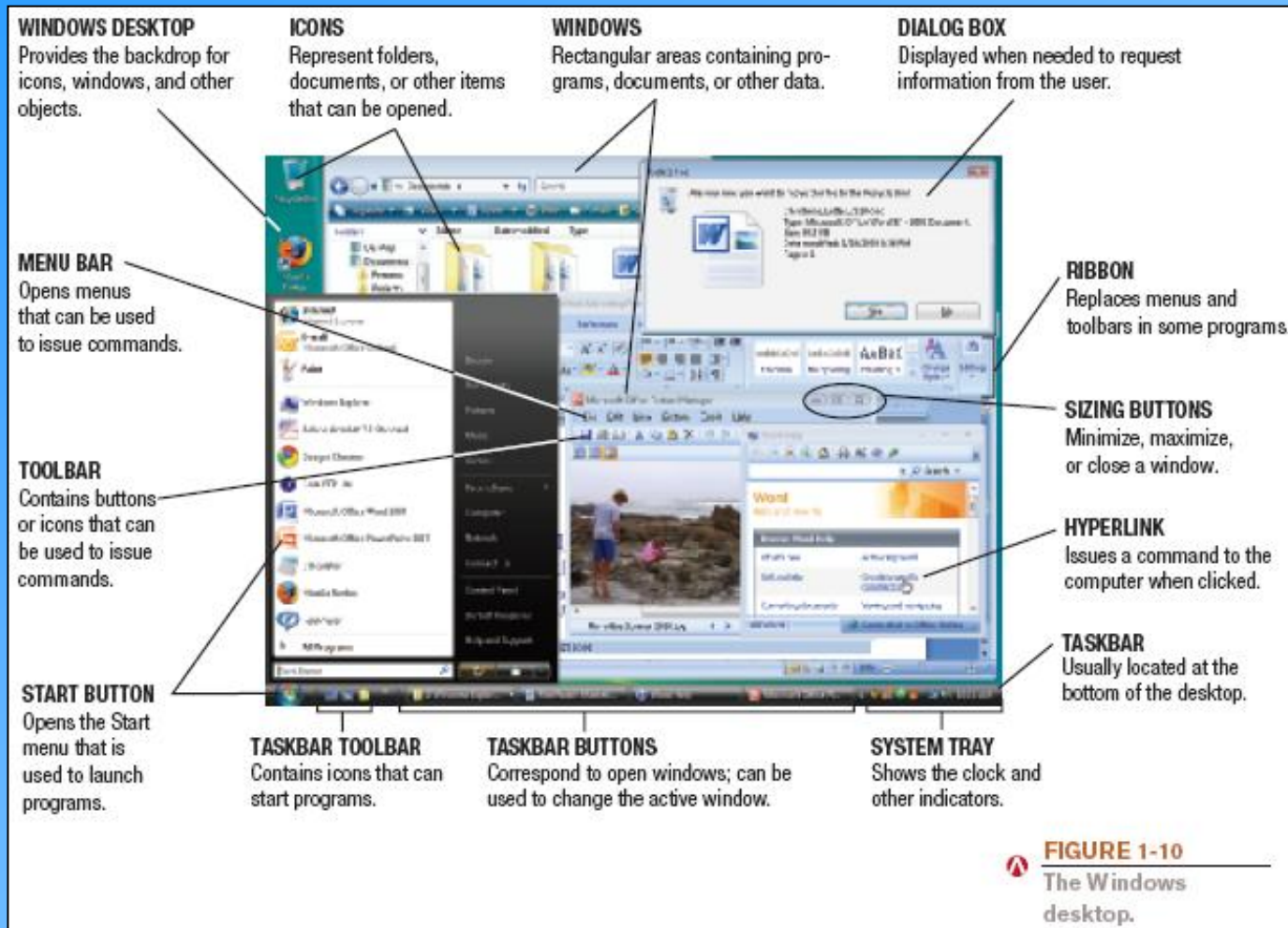
Hardware



Software

- Software: The programs or instructions used to tell the computer hardware what to do
 - System software: Operating system starts up the computer and controls its operation
 - Without OS computer cannot function
 - Boots the computer and launches programs at the user's direction
 - Most use a GUI to interact with the user via windows, icons, menus, buttons, etc.
 - Windows, Mac OS, Linux, etc.

Software



Application Software

- Application software: Performs specific tasks or applications
 - Creating letters, budgets, etc.
 - Managing inventory and customer databases
 - Editing photographs
 - Scheduling appointments
 - Viewing Web pages
 - Sending and receiving e-mail
 - Recording / playing CDs/DVDs
 - Designing homes
 - Playing games

Software



WORD PROCESSING PROGRAMS

Allow users to create written documents, such as reports, letters, and memos.



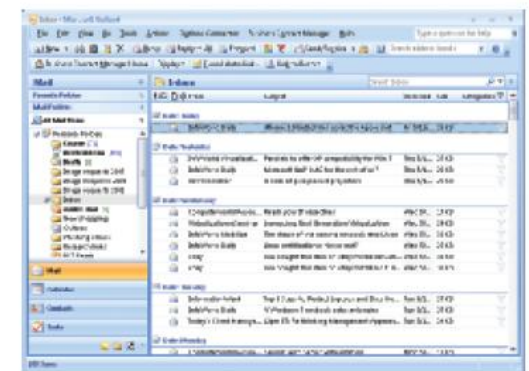
MULTIMEDIA PROGRAMS

Allow users to perform tasks, such as playing music or videos and transferring content to CDs and DVDs.



WEB BROWSERS

Allow users to view Web pages and other information located on the Internet.



E-MAIL PROGRAMS

Allow users to compose, send, receive, and manage electronic messages sent over the Internet or a private network.

FIGURE 1-11
Examples of application software.

Computer Users and Professionals

- Computer users (end users): People who use a computer to obtain information
- Computer professionals include:
 - Programmers
 - Systems analysts
 - Computer operations personnel
 - Security specialists

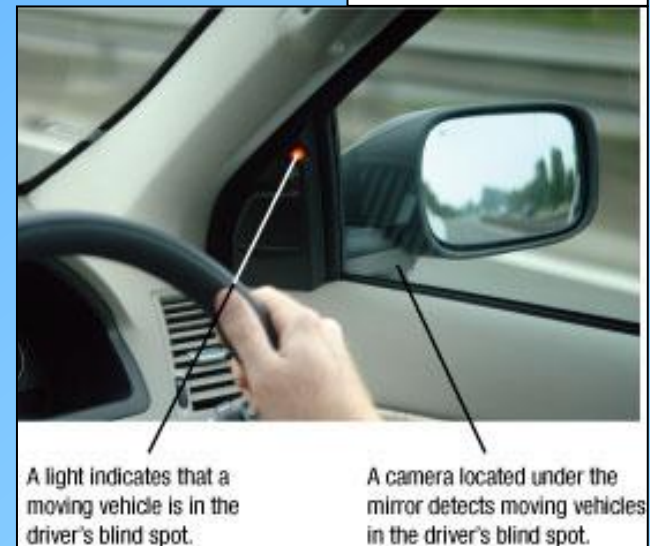
Computers to Fit Every Need

- Six basic categories of computers:
 - Embedded computers
 - Mobile devices
 - Personal computers
 - Midrange servers
 - Mainframe computers
 - Supercomputers

Embedded Computers

- Embedded computer: Embedded into a product and designed to perform specific tasks or functions for that product
- Cannot be used as general-purpose computers
- Often embedded into:
 - Household appliances
 - Thermostats
 - Cars

FIGURE 1-12
Embedded computers. This car's embedded computers control numerous features, such as notifying the driver when a car enters his or her blind spot.



Mobile Devices

- Mobile device: A very small device with some type of built-in computing or Internet capability
- Typically based on mobile phones
- Typically have small screens and keyboards
- Examples:
 - Smartphones
 - Handheld gaming devices
 - Portable digital media players



FIGURE 1-13
Smartphones. Most mobile devices today are based on the mobile phone.

Personal Computers (PCs)

- Personal computer: A small computer designed to be used by one person at a time
 - Also called a microcomputer
 - Range in size from desktop computers to UMPC's
- Desktop computers: Fit on or next to a desk
 - Can use tower case, desktop case, or all-in-one
 - Can be PC-compatible or Macintosh
 - Not designed to be portable



Portable Computers

- Notebook (laptop) computers: Typically use clamshell design
- Tablet computers: Can be slate tablets or convertible tablets
- Netbooks: Small notebooks; rapidly growing type of PC
- Ultra-mobile PCs (UMPCs): Handheld computers

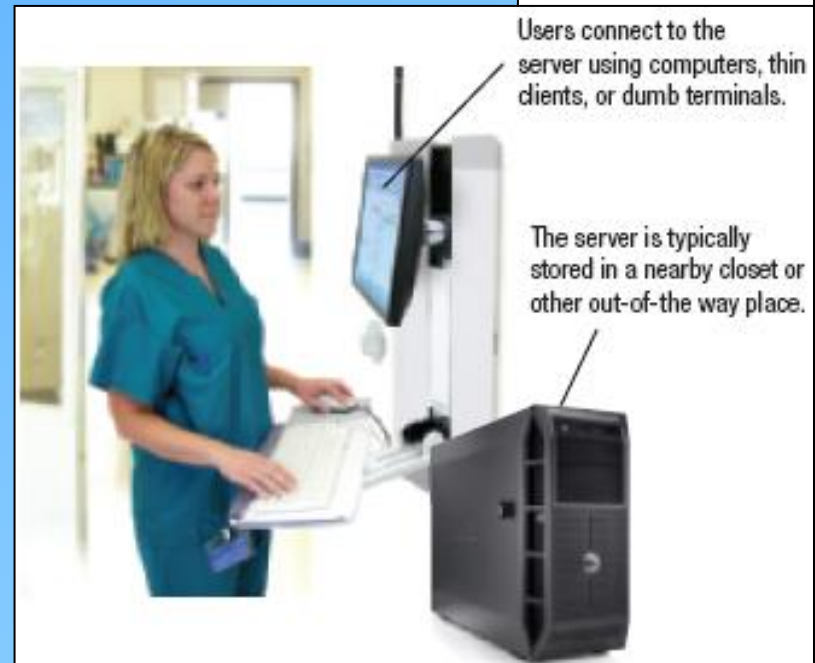


FIGURE 1-15
Portable computers.

Midrange Servers

- Midrange server: A medium-sized computer used to host programs and data for a small network
 - Users connect via a network with a computer, thin client.
 - May consist of a collection of individual circuit boards called blades
 - Virtualization: Creating virtual environments (often used to share a server for increased efficiency)

FIGURE 1-17
Midrange servers.
Midrange servers are used to host data and programs on a small network, such as a school computer lab or medical office network.



Mainframe Computers

- Mainframe computer: Powerful computer used by several large organizations to manage large amounts of centralized data
 - Standard choice for large organizations, hospitals, universities, large businesses, banks, government offices
 - Located in climate-controlled data centers and connected to the rest of the company computers via a network
 - Larger, more expensive, and more powerful than midrange servers
 - Usually operate 24 hours a day
 - Also called high-end servers or enterprise-class servers

Mainframe Computers

FIGURE 1-18
Mainframe
computers.



Supercomputers

- Supercomputer: Fastest, most expensive, most powerful type of computer
 - Generally run one program at a time, as fast as possible
 - Commonly built by connecting hundreds of smaller computers, supercomputing cluster
 - Used for space exploration, missile guidance, satellites, weather forecast, oil exploration, scientific research, complex Web sites, decision support systems, 3D applications, etc.

Supercomputers



FIGURE 1-19

The Roadrunner supercomputer.

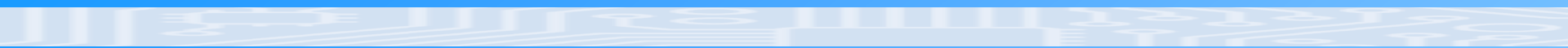
Supercomputers are used for specialized situations in which immense processing speed is required.

Quick Quiz

1. A tablet PC is an example of a(n) _____.
 - a. Desktop computer
 - b. Portable PC
 - c. Internet appliance
2. True or False: The terms mainframe computer and supercomputer are interchangeable; both refer to the largest, most powerful computers.
3. A smartphone is an example of a(n) _____.

Answers:

1) b; 2) False; 3) mobile device



Any Questions?